

Astrophysics SH2404

Radiation Transfer Lab

1 Preparatory Measures

You should study Rutten's compendium and solve the following problems. Approximate with an optically thin homogeneous medium, i.e. $S_\nu = \text{const.}$

1. A thin homogeneous slab of thickness Δs is irradiated from one side with a beam of intensity $I_\nu(s)$. What is the emergent intensity $I_\nu(s + \Delta s)$ on the other side if the emission coefficient in the slab is j_ν and if there is no extinction? Is the result also valid for a thick slab, with large Δs ?
2. A slab of thickness D is irradiated from one side with intensity $I_\nu(0)$. What is the emergent intensity $I_\nu(D)$ on the other side in the case of pure emission ($\alpha_\nu = 0$) and in the case of pure extinction ($j_\nu = 0$)? What are the results for a homogeneous slab?

3. Formally solve the transport equation, $dI_\nu/d\tau_\nu = S_\nu - I_\nu$! Do so by multiplying the equation with an integrating factor, getting an answer of the type $I(\tau_\nu(D)) = I(D) = \dots$, where D is the independent variable.

Note: τ designates optical length here and is often employed in optically thin applications. τ can, however, also stand for optical depth, which is frequently used in optically thick cases such as in stellar atmospheres. Both are applied in general radiative transfer discussions. Pay attention to which is referred to! How are they related?

4. A homogeneous medium contains particles that cause continuous emission j_ν^{cont} and extinction α_ν^{cont} at the frequency ν_0 , and also particles that cause bound-bound emission j_ν^{line} and extinction α_ν^{line} that is centered at ν_0 . The two corresponding source functions are the same: $S_\nu^{\text{cont}} = S_\nu^{\text{line}}$. Express the emergent intensity at the line frequency, for a beam that crosses the medium, in the above quantities for the following four cases:

- (a) $\tau_\nu(D) \gg 1$,
- (b) $\tau_\nu(D) < 1$ and $I_\nu(0) = 0$,
- (c) $\tau_\nu(D) < 1$ and $I_\nu(0) < S_\nu^{\text{total}}$,

(d) $\tau_\nu(D) < 1$ and $I_\nu(0) > S_\nu^{\text{total}}$!

What is in each case the character of the resulting spectral line (emission or absorption)?

2 Start up

You will need to run the Python programme `RadTransferII.ipynb` which is a Jupyter Notebook. This programme is on the course web. The program operates by widgets which will change the value of the corresponding variable. The idea is to play around with the interactive software and to eventually write a report on radiative transfer and line formation. Some questions are offered as a guide and a help for you to acquire the knowledge needed. When running the code you can print a plot simply by checking the "printplot" box. Uncheck it before you change the variables again to avoid saving too many plots.

3 Radiative Transport Through a Slab

By solving the radiation transport equation (equation of radiative transfer) for a homogeneous ($S_\nu = \text{const. w.r.t. } \tau$) medium one obtains the following equation:

$$I_\nu(D) = I_\nu(0)e^{-\tau_\nu(D)} + S_\nu^{\text{tot}}(1 - e^{-\tau_\nu(D)}),$$

where τ is the optical path, and D is the extension of the medium. In this part of the computer exercise an optically thin slab is modelled.

The four variables of interest here are the incident intensity $I(0)$, the source function S , the continuum optical depth τ_c and the optical depth in the spectral line τ_l . By varying all these parameters, different cases of spectral line formation in a homogeneous slab can be illustrated. Explain the two different optical depths! Does the sum of the two have a meaning?

First play around and try to cover all cases! Draw relevant pictures and make notes for the report!

1. Start off systematically by, for instance, setting I_0 and τ_c to zero. What does it mean to set τ_c to zero? What happens if you change τ_l ? Write down interesting values and what you observe. Give explanations! When does the line saturate? Why does it not saturate at $\tau_l = 1$?
2. Choose a $0 < \tau_l < 1$ and increase I_0 ! What happens when I_0 is greater than the source function?
3. Is there any fundamental difference between a saturated absorption line and a saturated emission line? Give an example of values of the different variables for the two cases!

4. What does τ_c and τ_l (line only) equal to zero show?
5. What does $\tau_c \neq 0$ propose in the case of $I_0 < S$ and $I_0 > S$? Choose some values of I_0 , S and τ_c . Do some calculations to ensure that the theory is correct!
6. What shape does an unsaturated line have? Why is that? Try to give an explanation!
7. Does $\tau_c = \tau_l$ imply anything?
8. Can you illustrate the optically thick case?
9. For the case of $\tau_c = 0$ and $\tau_l = 1$, does the line center reach the level of S_ν ? Why is this not the case? What happens if you change the damping parameter ($\Gamma_u/4\pi \equiv \text{damping}$) to be very small. Explain! What does a small damping parameter signify?

4 Report

A (short) written report is required. It should contain a description of what you have been investigating and results achieved from running the programme, both in text and pictures. Also, you should compose a maximum of five sentences, describing how either an absorption or an emission line is formed! In order to pass, you should have taken an active part in doing the computer lab and also have handed in a satisfactory report.